

Question number	Answer	Additional guidance	Mark
3(a)	An explanation that includes the following points: • because it {destroys / weakens} the immune system (1) • by {destroying / decreasing the number of} T helper cells (1) • so {B cells / T killer cells} cannot be activated (1) • credit a consequence of this (1)	ACCEPT immunity ACCEPT CD4 cells ACCEPT {humoral / cell-mediated} immunity cannot be initiated e.g fewer antibodies produced (by plasma cells), less opsonisation, fewer infected host cells destroyed IGNORE phagocytosis unless description of opsonisation DO NOT ACCEPT viruses killed NB max 3 marks if no context	(4)

Question number	Answer	Additional guidance	Mark
3(b)(i)	An explanation that includes the following points: - because if reverse transcriptase is inhibited, a DNA copy of the (viral) RNA cannot be made (1) - because if integrase is inhibited, this DNA cannot be incorporated into the host cell DNA (1)	ACCEPT (viral) DNA cannot be made (viral) RNA cannot be transcribed <u>into DNA</u> DO NOT ACCEPT RNA cannot be made into DNA ACCEPT provirus cannot be formed NB both reasons given but not linked to each enzyme = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	An answer that includes the following points: - to increase the likelihood of the drugs {being effective / destroying the virus} (1) - credit an example of why they may not be effective on their own (1) - maybe used to treat other (viral) infections (1)	ACCEPT a description of how this could happen e.g.targetetting another site (on the virus) DO NOT ACCEPT kill the virus e.g. one type of inhibitor may not penetrate all the cells virus may have mutated and become resistant other drugs might prevent the attachment of HIV to {host cells / T helper cells / CD4}	(2)

Question number	Answer	Additional guidance	Mark
3(b)(iii)	An answer that includes the following points: • because the drug treatment does not completely remove the virus (1) • therefore {viral replication can occur / activation of provirus} (1) • new viral particles burst out of {host / T helper} cells (1)	ACCEPT enter the lytic cycle DO NOT ACCEPT if linked to incorporation of DNA into host DNA	(2)